

SHARDUL HIROLIKAR

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EDUCATION

Vishwakarma Institute of Technology, Pune

2024 – 2028

B.Tech, Computer Science (Artificial Intelligence) | CGPA: **9.22/10**

Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Foundations of AI & ML, Computer Architecture

ACHIEVEMENTS

- **Indian Patent Application Published** (App. No. 202521113731) for a Smart Retrofit Fan Control System using **Fuzzy Logic-Based Adaptive Control** — designed the control logic, simulated adaptive thresholds, and validated a working hardware prototype.
- **National Semi-Finalist, Arbitrage Arena 2026 (IISc Bangalore)** — built an **LSTM** financial time-series forecasting model with custom feature engineering to backtest trading strategies against real market data.

TECHNICAL SKILLS

Languages: C++, Python, TypeScript, JavaScript, SQL, C

Frameworks: Next.js, React, Node.js, Express, Flask

Databases: PostgreSQL/Supabase, MySQL, Firebase

ML/Data: NumPy, Pandas, CLIP, LSTM

Concepts: DSA, OOP, DBMS, OS, Computer Networks, REST APIs

PROJECTS

AI-Driven Water Leakage Detection & Predictive Maintenance (*CDAC Pune, Ongoing*)

Industry-collaborative project · Machine Learning · Predictive Maintenance

- One of **4 students selected** for an **industry-collaborative** initiative with **CDAC Pune**, developing **ML-driven leak-detection** and predictive-maintenance models for citywide water distribution networks.

AceMyInterview – AI-Powered Mock Interview Platform ([GitHub](#), [Live demo](#))

Next.js · TypeScript · Supabase · Gemini API

- Designed and shipped a full-stack AI mock-interview platform end-to-end on **Next.js** and **Supabase**, covering auth, real-time speech interaction, AI evaluation, and analytics.
- Architected a **single-pass AI evaluation pipeline** that consolidates multi-question data into one **Gemini API** call, cutting token overhead **40%+** and reducing response latency.
- Modelled a normalized **PostgreSQL** schema (7+ tables) enforced with **Supabase Auth** and **Row-Level Security (RLS)** for strict multi-tenant data isolation.
- Delivered a hands-free interview flow using native browser **Speech-to-Text/Text-to-Speech** with low-latency state recovery.

Multimodal Hybrid Vector Database ([GitHub](#))

Python · CLIP Embeddings · HNSW

- Engineered a **hybrid vector database from scratch** in pure Python for multimodal ingestion of text, PDFs, and images, combining embedding search with exact metadata filtering.
- Built an **HNSW (Hierarchical Navigable Small World)** index from scratch for approximate nearest-neighbour search, plus insertion, similarity search, deletion, and persistence layers.
- Unified text and image retrieval with **CLIP embeddings** and cut repeated-query latency using a custom **LRU cache**.
- Benchmarked multimodal retrieval at **0.085 ms** end-to-end latency, with **0.072 ms** HNSW query latency and **80% Recall@5**.

Real-Time Cloth Physics Engine ([GitHub](#), [Live demo](#))

JavaScript · HTML5 Canvas

- Engineered a **Verlet-integration** physics engine handling **3,200+ particles** and 7,200+ constraints while sustaining **~800 FPS** (sub-2.5ms frame times) under heavy stress testing.
- Eliminated runtime allocation and GC overhead with **JavaScript TypedArrays**, and built a custom **4-pass Gauss-Seidel constraint solver** enabling real-time dynamic mesh cutting.